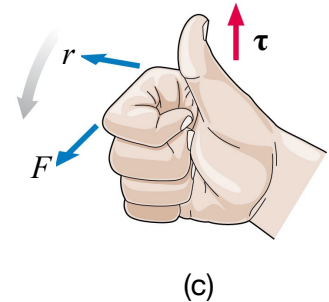
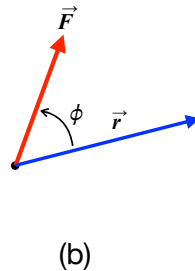
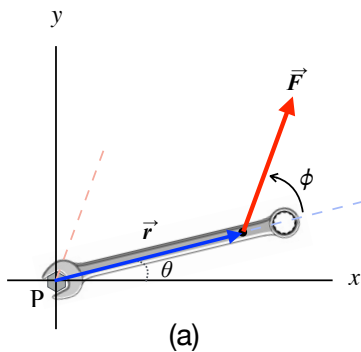


Calculating Torque

CONCEPT: Torque (“moment of force” in engineering) is the rotational counterpart of force. It is a measure of the ability of a force to cause or change the rotation of an object.

The torque $\vec{\tau}$ about a point P, called the pivot or fulcrum, is given by the cross product between the position vector $\vec{r}$ and the force vector $\vec{F}$. In vector notation, this is written as:

$$\vec{\tau} = \vec{r} \times \vec{F}.$$



MAGNITUDE: The magnitude of the torque about a pivot point P is given by:

$$\tau = r F \sin \phi$$

where:

- r is the magnitude of the position vector $\vec{r}$, it is the distance from the pivot to the point of application of the force (see Fig. (a));
- F is the magnitude of the force $\vec{F}$; and
- ϕ is the angle between the tails of vectors $\vec{r}$ and $\vec{F}$ (see Fig. (b)).

DIRECTION: The torque vector is perpendicular to the plane defined by vectors $\vec{r}$ and $\vec{F}$. In the example of Fig. (a), the torque points along the +z-axis, that is, out of the page.

Use the right-hand rule to determine directions, as shown in Fig. (c). By convention, the direction of the torque is considered:

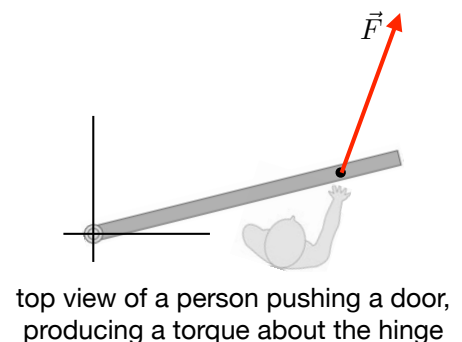
- ⊙ **positive** (+z-axis) if it causes a counterclockwise (ccw) rotation; and
- ⊗ **negative** (-z-axis) if it causes a clockwise (cw) rotation.

VISUALIZING TORQUE:

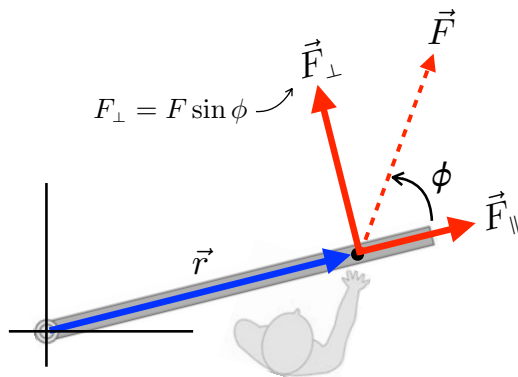
Torque requires a perpendicular component between the vectors $\vec{r}$ and $\vec{F}$, as indicated by the $\sin \phi$ term.

There are two ways of visualizing it:

1. due to the component of the force perpendicular to $\vec{r}$;
2. due to the component of $\vec{r}$ perpendicular to the force.



METHOD 1: $F_{\perp}$
(F perpendicular to r)

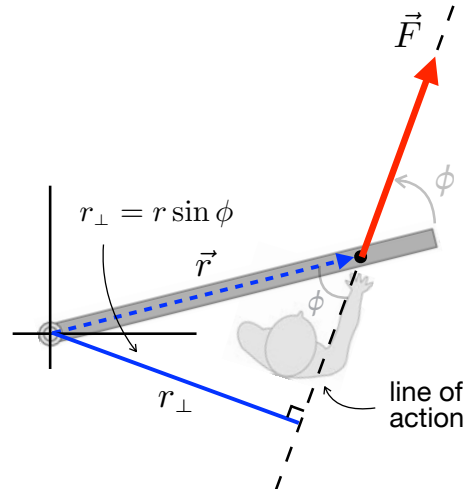


- identify the vector $\vec{r}$, from the pivot to the point of application of the force;
- identify $\vec{F}_{\perp}$, the component of the force perpendicular to $\vec{r}$ (while $\vec{F}_{\parallel}$ is the component parallel to $\vec{r}$).

Torque due to distance r multiplied by F -perpendicular ($F_{\perp}$):

$$\tau = r (F_{\perp})$$

METHOD 2: $r_{\perp}$
(r perpendicular to F)



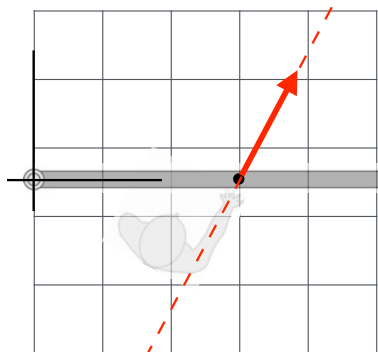
- identify the line of action, the imaginary line along which the force acts;
- identify the moment or lever arm $r_{\perp}$, which is the minimum distance from the pivot to the line of action.

Torque due to moment arm r -perpendicular ($r_{\perp}$) multiplied by the force F :

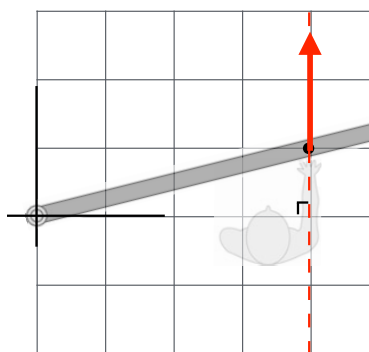
$$\tau = (r_{\perp}) F$$

Note:

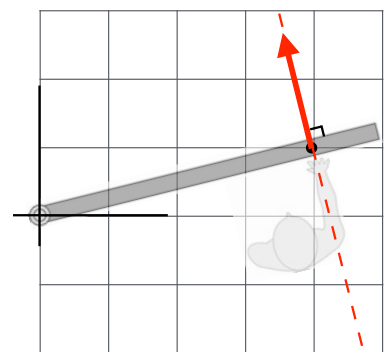
- Both methods of visualizing torque are equivalent and provide the same result.
- Select a method based on whether you know r or $r_{\perp}$ (or equivalently, F or $F_{\perp}$).
- Use the geometry of the problem to determine or calculate the components $F_{\perp}$ or $r_{\perp}$.
- Recall that when the line of action of a force crosses the pivot, the torque exerted by that force is zero.



r is known, $r = 3$ units
definition/method 1 suitable



$r_{\perp}$ is known, $r_{\perp} = 4$ units
method 2 suitable



$r = r_{\perp}$
any method is suitable